Exploring single-cell images and profiles together with CytoDataFrame

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I. Why separate profiles and images?

Figure 1: ***Feature tables and images often live in separate tools, making it hard to build intuition about single cells. CytoDataFrame keeps them together in one interactive table inside your notebook.***

High-content screens produce **millions of rows of features** and **many images**. Jumping between a DataFrame and an image viewer slows exploration and hides issues like segmentation errors or odd phenotypes. **CytoDataFrame** puts an **image thumbnail** right next to the row's features so you can **see** how features relate to images.

II. CytoDataFrame for integrated analysis

CytoDataFrame

Figure 2: ***Images and profile data may be mixed using Pandas idioms through Jupyter interfaces to create a unified experience for bioimage analysts and data scientists.***

CytoDataFrame is a small package built on top of `pandas.DataFrame` that embeds **inline images** (cells, nuclei, or other compartments) alongside per-object features within **Jupyter** notebook interfaces. Using CytoDataFrame enables you to:

- **Keep context at a glance** - Each row shows the features *and* the object.
- **Filter and explore** - Subset outliers, phenotypes, or QC flags and see what they look like immediately.
- **Stay in your workflow** - It behaves like a DataFrame; your plotting, modeling, and exports still work.
- **Fantastic for integrations** - Tight integration with packages such as **coSMicQC** returns CytoDataFrames so you can act on QC signals with visuals.

III. Using CytoDataFrame in Python

a) Install

```
# install CytoDataFrame from PyPI
pip install cytodataframe
```

b) View images with your profiles

```
from cytodataframe import CytoDataFrame

# Load features and images together.
CytoDataFrame(
    data="BR00117006.parquet",
    data_context_dir="images/orig",
)[[
        "Metadata_ImageNumber",
        "Cells_Number_Object_Number",
        "Image_FileName_OrigAGP",
        "Image_FileName_OrigDNA",
        "Image_FileName_OrigRNA",
  ]]
```

Image adjustment: 50

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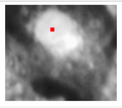
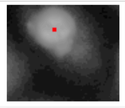
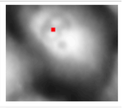
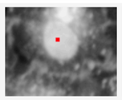
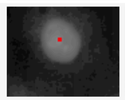
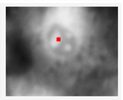
0	1	1			
1	1	2			

c) Show segmented objects with profiles

```
# load outlines to show segmented images
CytoDataFrame(
    data="BR00117006.parquet",
    data_context_dir="images/orig",
    data_outline_context_dir=f"images/outlines",
)[[
        "Metadata_ImageNumber",
        "Cells_Number_Object_Number",
        "Image_FileName_OrigAGP",
        "Image_FileName_OrigDNA",
        "Image_FileName_OrigRNA",
  ]]
```

Image adjustment: 50

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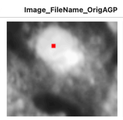
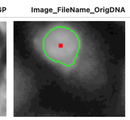
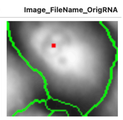
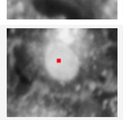
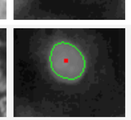
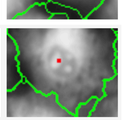
0	1	1			
1	1	2			

d) Change display configuration for images

```
# change the color of your outlines (and more!)
CytoDataFrame(
    data="BR00117006.parquet",
    data_context_dir="images/orig",
    data_outline_context_dir=f"images/outlines",
    display_options={
      "outline_color": (200, 100, 255)
    },
)[[
        "Metadata_ImageNumber",
        "Cells_Number_Object_Number",
        "Image_FileName_OrigAGP",
        "Image_FileName_OrigDNA",
        "Image_FileName_OrigRNA",
  ]]
```

Image adjustment: 50

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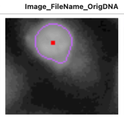
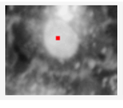
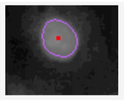
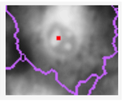
0	1	1			
1	1	2			

e) Add scale bars for objects

```
# add scale bars using reproducible configuration
CytoDataFrame(
    data="BR00117006.parquet",
    data_context_dir="images/orig",
    data_outline_context_dir=f"images/outlines",
    display_options={
        "pixel_per_um": 0.1550,
        "scale_bar": {"length_um": 100,
            "location": "lower right",
            "color": (255, 255, 255),
            "thickness_px": 2, "margin_px": 5,},
    },
)[[
        "Metadata_ImageNumber",
        "Cells_Number_Object_Number",
        "Image_FileName_OrigAGP",
        "Image_FileName_OrigDNA",
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  ]]
```

Image adjustment: 50

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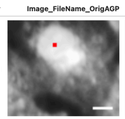
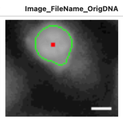
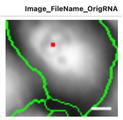
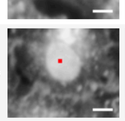
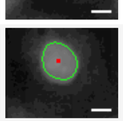
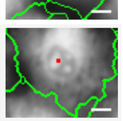
0	1	1			
1	1	2			

IV. coSMicQC leverages CytoDataFrames

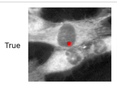
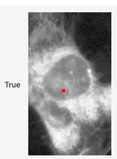
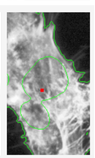
Cytomining

coSMic QC Integrates with CytoDataFrame

```
import cosmicqc

# find and label outliers for single-cell data
# and return a cytodataframe for use.
cosmicqc.analyze.label_outliers(
  df=CytoDataFrame(
      data="single-cell-profiles.parquet,
      data_context_dir="images/orig",
      data_outline_context_dir=f"images/outlines",
  ),
  include_threshold_scores=True,
)[[
        "Metadata_ImageNumber",
        "Metadata_Cells_Number_Object_Number",
        "cqc.large_nuclei.is_outlier",
        "Image_FileName_GFP",
        "Image_FileName_RFP",
        "Image_FileName_DAPI",
  ]]
```

Image adjustment: 50

	Metadata_ImageNumber	Metadata_Cells_Number_Object_Number	cqc.large_nuclei.is_outlier	Image_FileName_GFP	Image_FileName_RFP
699	50	2	True		
1557	113	10	True		

coSMicQC functions can return CytoDataFrames, letting you pair quality control flags with images to make immediate, visual decisions and improve your profiling outcomes.

V. Acknowledgements

We thank those who have inspired, contributed, or helped support CytoDataFrame and the Cytomining Ecosystem:

- Open source science from the teams behind **CellProfiler**
- Members of the **Way Lab** at the University of Colorado Anschutz Medical Campus
- **Department of Biomedical Informatics** within the School of Medicine at the University of Colorado Anschutz Medical Campus